

A Hybrid RAG System with Graph Based Context Retrieval

Our Project is the Graph RAG Powered Academic Assistant, an AI based tool that augments Retrieval Augmented Generation (RAG) with citation graph analysis. It ingests research papers, builds citation-aware networks, and provides natural language answers with traceable citations. Key features include dual retrieval (semantic + citation graph), citation trail visualization, and explainability modules.

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1. Abstract

Academic research is increasingly dependent on the ability to navigate, interpret, and connect information across multiple scholarly papers. Despite advancements in Natural Language Processing (NLP), students and early researchers continue to struggle with understanding complex academic texts, especially when tracing the evolution of ideas through citations. Large Language Models (LLMs) can assist with natural language understanding, but their responses often lack grounding, transparency, and traceability that are essential in academic contexts. Retrieval-Augmented Generation (RAG) offers partial solutions but falls short when dealing with citation-heavy domains because it relies solely on semantic similarity.

To address these limitations, this capstone project presents the Graph RAG Powered Academic Assistant, a hybrid retrieval pipeline that integrates semantic vector search with citation graph traversal. The system parses academic PDFs, extracts citations, constructs a directed citation graph, and performs dual retrieval to produce answers that are both contextually relevant and academically verifiable. The assistant includes a comprehensive explainability module capable of displaying citation trails, chunk level attribution, and reasoning paths. A user-friendly streamlit-based interface enables uploading papers, interacting with the model, and viewing visualizations.

Evaluation of the system using a benchmark of 10 academic papers relied on a general qualitative evaluation approach, given that the primary focus of this project was architectural design rather than numerical benchmarking. The evaluation emphasized relevance of retrieved chunks, citation awareness, grounding of LLM outputs, reduction of hallucinations, and transparency of reasoning. Phoenix-based tracing further validated the retrieval-to-reasoning flow by inspecting context passed to the model and verifying that generated answers remained grounded in retrieved evidence. These observations collectively highlight the value of incorporating structural academic relationships such as citation graphs into retrieval workflows. The project demonstrates a transparent, trustworthy, and scalable solution for academic question answering and positions Graph-RAG as a promising direction for future scholarly AI systems.

2. Introduction

2.1 Background

Academic literature is foundational to scientific progress yet navigating it effectively remains a significant challenge. Papers are laden with references, dense writing styles, and conceptual dependencies that require a strong understanding of how ideas evolve across publications. While NLP tools have improved the accessibility of academic content, many still fail to capture the structural context that citations provide. Most RAG systems rely on chunk-based semantic retrieval, ignoring how papers influence each other over time.

2.2 Motivation

Students and researchers typically rely on manual citation tracing moving between reference lists, linked papers, and review articles. This is time-consuming and often overwhelming. A system that not only retrieves relevant information but also explains why certain papers matter and how they are connected can significantly enhance learning outcomes. The motivation behind this project is to build an AI assistant that:

- Retrieves information based on meaning and citation relevance.
- Provides transparent, verifiable answers.
- Supports academic integrity by showing how conclusions are derived.

2.3 Limitations of Existing Tools

This section summarizes the key limitations of current academic research tools and highlights gaps addressed by our system.

- **Traditional RAG:** Only semantic; easily hallucinated.
- **Connected Papers:** Visualizes citation networks but lacks QA capabilities.

- **Vanilla LLMs:** Cannot cite accurately without structured retrieval.

2.4 Objectives

The objectives of this project are outlined below, defining the core goals that shape the system's development.

- Build a hybrid RAG engine combining semantic retrieval with citation graph traversal.
- Produce explainable answers backed by verifiable sources.
- Generate visual insights to help users understand research lineages.
- Reduce hallucinations and improve academic trustworthiness.

2.5 Contributions

This project offers several contributions:

- A working Graph-RAG engine capable of dual retrieval.
- A citation graph builder using OpenAlex.
- An explainability subsystem with citation trails and Phoenix tracing.
- A modular, local, privacy-preserving system architecture.
- A streamlined UI for student and researcher use.

3. Problem Statement & Significance

Traditional RAG systems lack the ability to incorporate structural relationships such as citation networks, leading to incomplete or inaccurate academic QA. The problem addressed is how to augment RAG with citation graphs to improve the reliability, transparency, and learning value of AI-generated academic answers.

Significance: Making AI assistants explainable is not just a technical enhancement but a trust building requirement in academic and professional settings. Our assistant improves user trust and comprehension by:

- Linking concepts through citation graphs

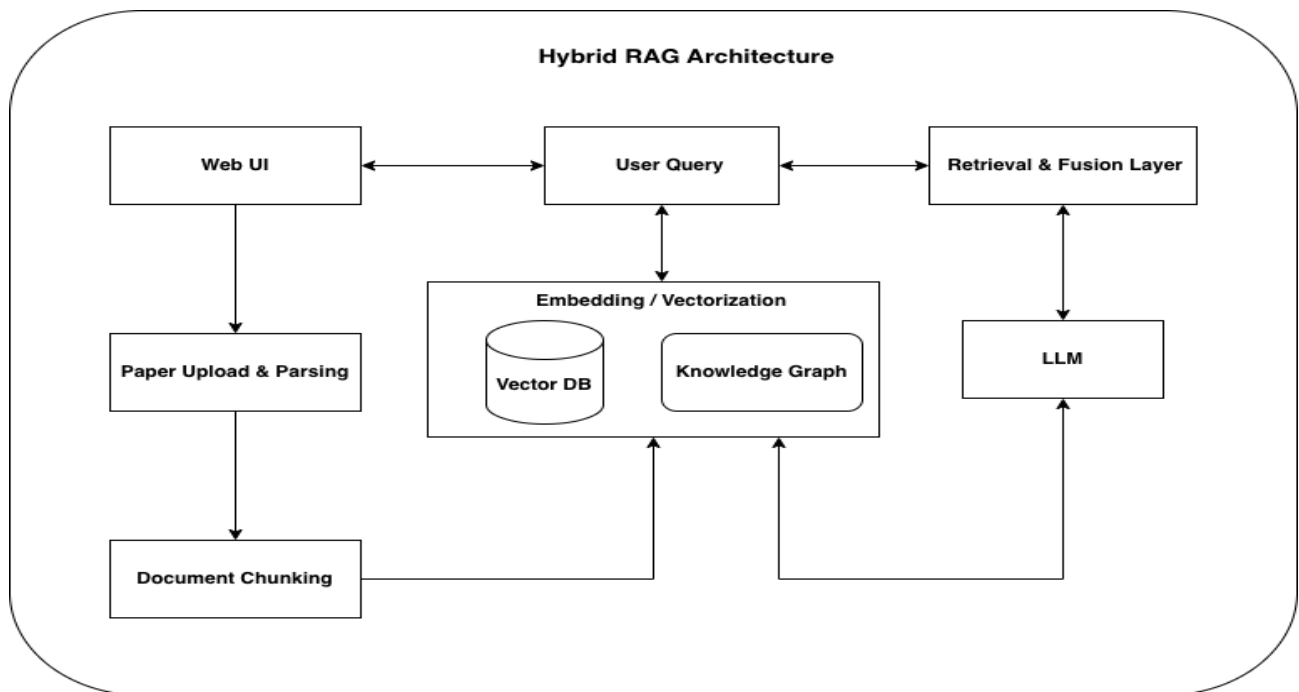
- Providing visible paths for answers
- Reducing hallucinated or ungrounded answers

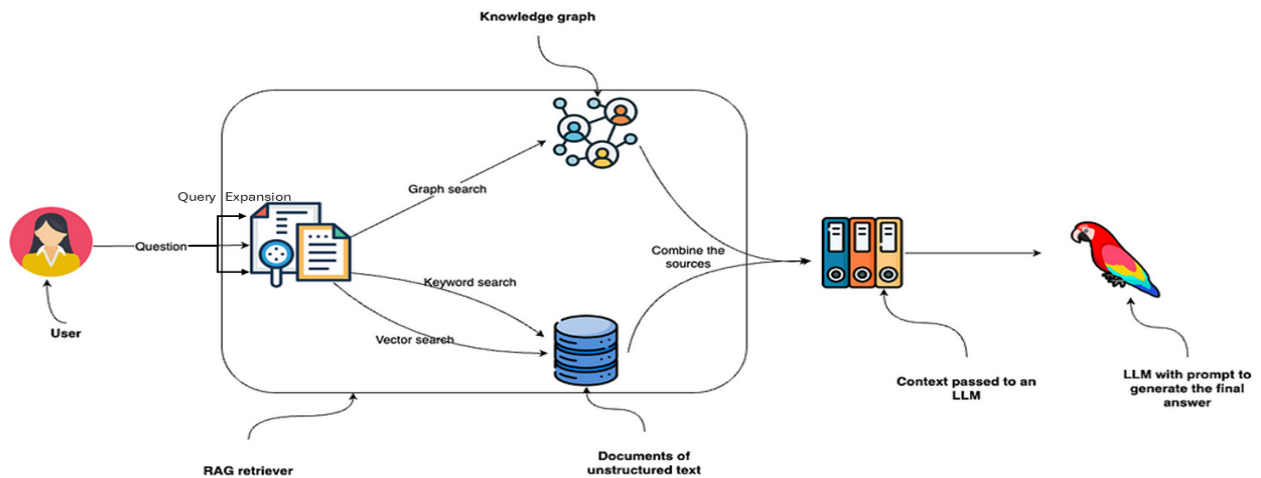
This project empowers researchers, students, and educators to interact with academic literature in a guided and informed manner.

4. System Overview & Architecture

4.1 Overview

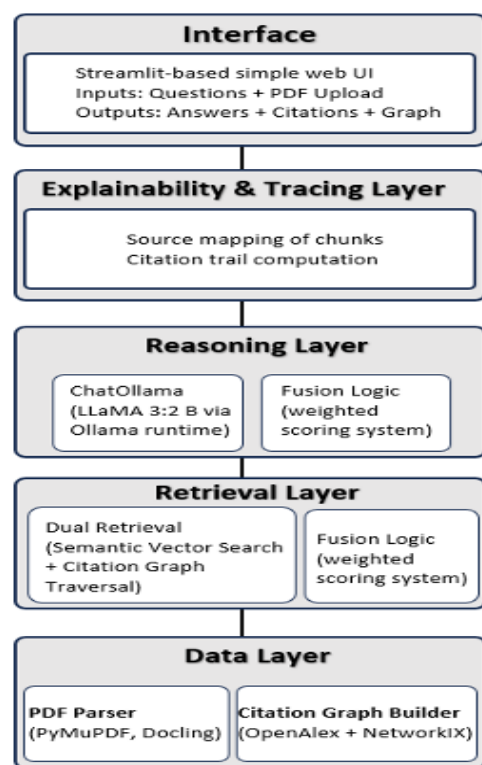
The system follows a modular Hybrid RAG Architecture designed to integrate semantic vector retrieval with citation-graph based reasoning. The flow begins with the Web UI, where users upload research papers and submit natural language queries. Uploaded PDFs are processed through the Paper Upload & Parsing module, which extracts text, metadata, and references. The extracted content is then forwarded to the Document Chunking component, where the text is segmented into semantically meaningful chunks optimized for retrieval.





4.2 Layered Architecture

The system is structured into five modular layers, each responsible for a distinct portion of the Graph RAG workflow. This layered approach improves maintainability, scalability, and clarity of data flow throughout the system.



4.3 Data Layer

The system will accept academic papers in PDF format and automatically parse them into semantically meaningful text chunks with metadata such as authors, year, and references. This ensures that each piece of content retains context while being optimized for retrieval. By structuring papers into smaller, annotated segments, the assistant can provide more accurate and focused answers to user queries. A citation graph will be constructed using external APIs such as OpenAlex, where nodes represent papers and edges represent citation relationships. This feature allows users to see how ideas are connected across papers, tracing the flow of knowledge through multiple generations of research. The interactive visualization of this graph will make it easier for students and researchers to follow academic lineages.

4.4 Retrieval & Reasoning Layer

Unlike traditional RAG systems that rely only on semantic similarity, our assistant combines vector based semantic retrieval with citation graph traversal. This dual approach ensures both contextual relevance and citation supported accuracy. The fusion of results reduces hallucinations while improving trust in AI generated answers by grounding them in verifiable sources.

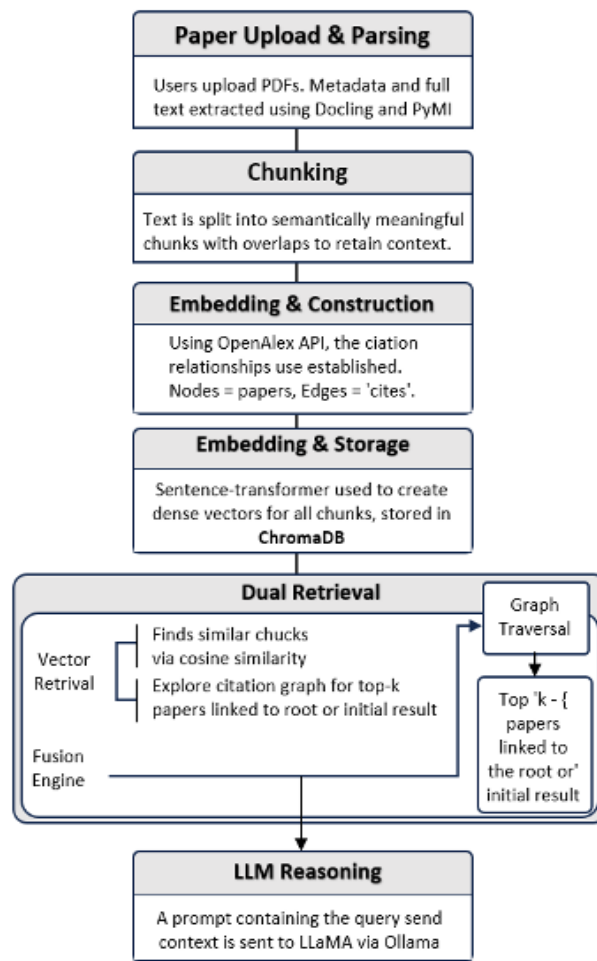
4.5 Explainability Layer

This module enhances transparency by showing users how the system arrived at an answer. Features include citation trails (e.g., Vaswani → Bahdanau → Luong), chunk-level attribution, and confidence scoring. By making reasoning paths explicit, students can verify claims directly in the source material, ensuring academic integrity and building trust in the assistant's outputs. LLM Request Tracing: Phoenix captures every LLM call along with inputs, outputs, and execution metadata.

4.6 Interface Layer

A simple, interactive UI web interface. Users will be able to upload research papers, ask natural language questions, and view answers paired with citations and graphs. The UI is designed for accessibility, making it suitable for academic settings such as classrooms, labs, and independent study.

5. Methodology



5.1 End-to-End Workflow

1. **PDF Upload:** The user uploads a research paper in PDF format. The system extracts raw text, metadata and reference sections using parsing libraries. This step ensures all structural components of the document are captured for downstream processing.

2. **Chunking:** The extracted text is segmented into overlapping chunks. Overlaps preserve contextual continuity across boundaries, enabling the retrieval engine to locate relevant information even when it spans sections.
3. **Embedding:** Each chunk is transformed into a dense vector embedding using a sentence transformer model. These embeddings represent semantic meaning, allowing efficient similarity-based retrieval within the vector database.
4. **Graph Construction:** Using metadata and external APIs such as OpenAlex, the system builds a citation graph that maps directional relationships between papers. Nodes represent papers, while edges represent citation links, forming the structural backbone for graph-based retrieval.
5. **Dual Retrieval:** When a query is submitted, the system performs both semantic vector search and citation-graph traversal. Vector retrieval finds semantically similar chunks, while graph retrieval identifies conceptually or historically related papers.
6. **Fusion:** A weighted scoring engine merges outputs from the two retrieval strategies. The fusion system balances semantic similarity, citation relevance, and hop distance to produce a unified list of highly relevant passages.
7. **LLM Reasoning:** The fused retrieval results are injected into a tailored prompt template and passed to the LLM. The model synthesizes a coherent, contextually grounded answer based on retrieved evidence.
8. **Explainability:** The system generates citation trails, chunk-level attributions, and Phoenix traces. This ensures transparency by showing precisely which documents and reasoning paths contributed to the final answer.

5.2 Prompt Design

Prompts include context passages and instructions to cite specific chunk IDs.

```
prompt = ChatPromptTemplate.from_template(  
    "Generate up to 4 concise paraphrases or closely related query variants for: '{q}'.\n"  
    "Return a JSON array of strings only."  
)
```

```
prompt = ChatPromptTemplate.from_messages([  
    ("system", (  
        "You are a precise research assistant. Answer strictly using the provided context. "  
        "If the context is insufficient, say you don't know.\n\n"  
        "Requirements:\n"  
        "- Prefer concise bullet points (3-5).\n"  
        "- Include inline citations using the form [Title, Year] when available.\n"  
        "- Do not fabricate facts or citations.\n\n"  
        "<context>\n{context}\n</context>"  
    )),  
    ("human", "{question}")  
)
```

6. Implementation

6.1 Tools and Frameworks

- Python 3.10+
- ChromaDB
- NetworkX
- Streamlit
- LangChain
- LangChain-Ollama
- LangChain-Docling
- SentenceTransformers
- Phoenix + OpenInference (Tracing)

6.2 Features

Controls

Upload PDF Papers

Drag and drop files here
Limit 200MB per file • PDF

Browse files

NIPS-2017-attention-is-a...
0.5MB

Processing uploaded PDFs...

Instructions

- Upload one or more PDF papers in the sidebar.
- Enter a question related to the papers (e.g., "What is scaled dot-product attention?").
- Click **Submit Query** to get an answer with explanations.
- View the interactive citation graph below, showing relationships between papers.

Clear Papers and Reset

RAG Citation Graph Assistant

Upload academic papers and ask questions to get answers with citation context and explanations.

Loaded Papers:

- NIPS-2017-attention-is-all-you-need-Paper.pdf

Enter your question:

what is the attention concept in the paper

Submit Query

Processing query...

Answer

In the context of deep learning and natural language processing, attention refers to a mechanism that allows a model to focus on specific parts of an input sequence (such as words or tokens) when generating an output. This is particularly useful for tasks like machine translation, question answering, and text summarization.

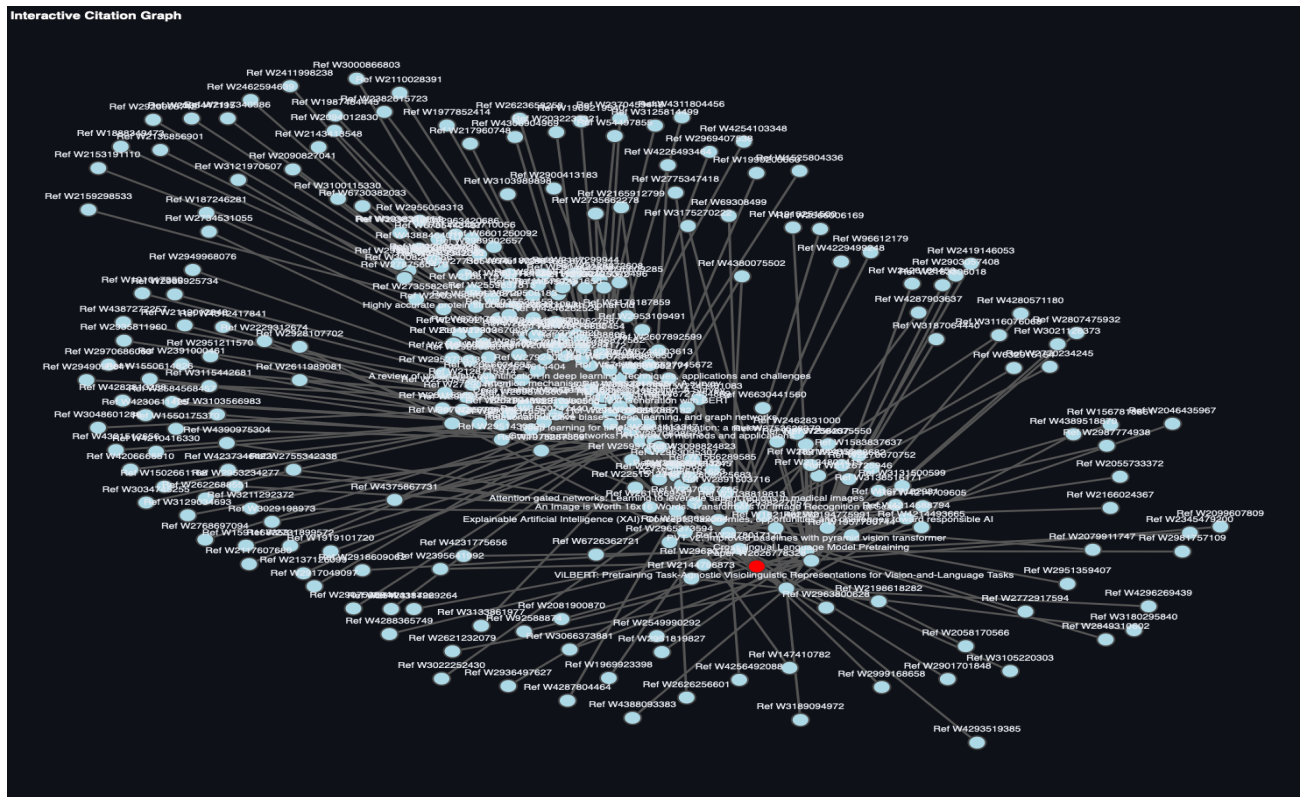
The basic idea behind attention is to weigh the importance of each input element relative to others, so that the model can selectively concentrate on the most relevant information. This is achieved by computing a weighted sum of the input elements, where the weights represent the model's attention towards each element.

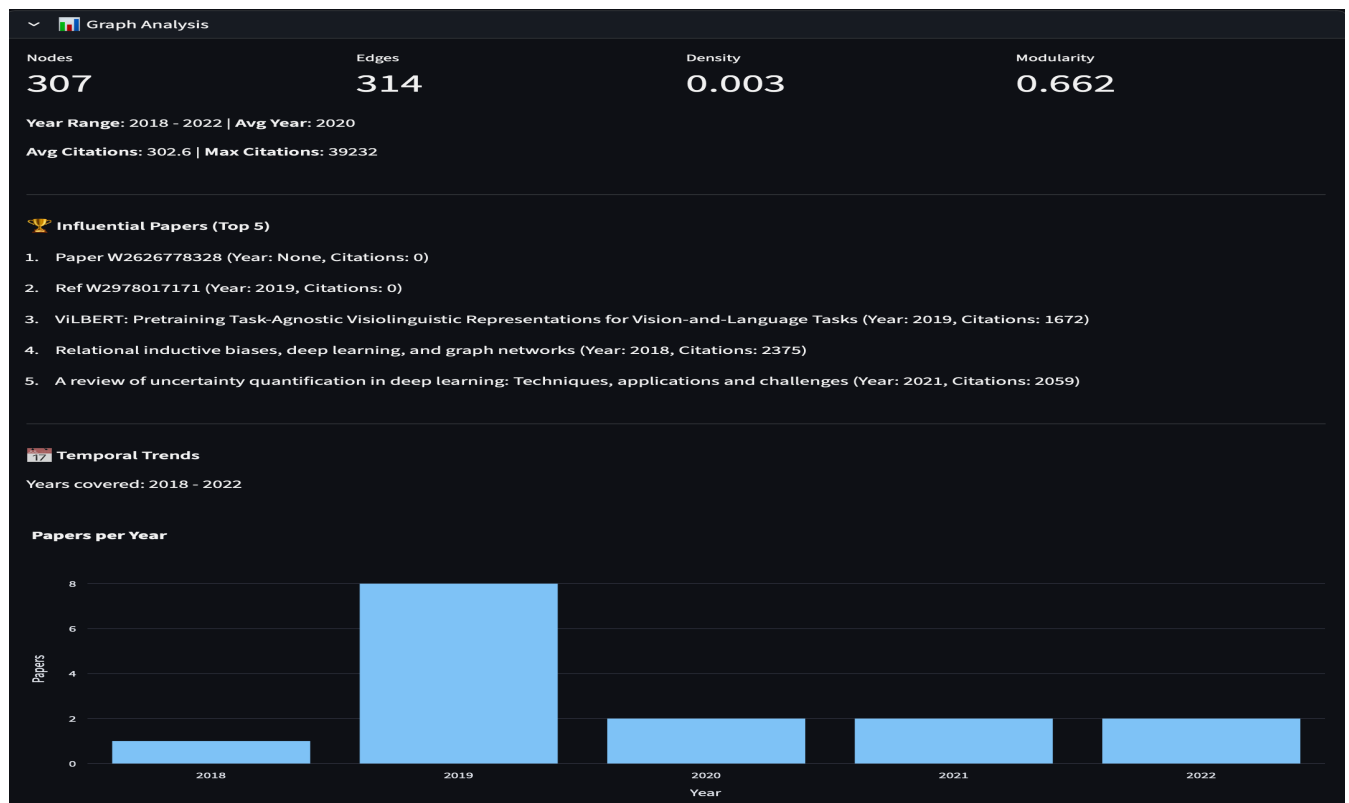
In the context of neural networks, attention mechanisms are often used in conjunction with recurrent or transformer architectures to improve performance on tasks that require sequential processing, such as language modeling and machine translation.

Explanation

Content: - [46] B. A. Olshausen, C. H. Anderson, and D. C. V. Essen, "A neurobiological model of visual attention and invariant pattern recognition based on dynamic routing of information," *Journal of Neuroscience*, 1993.

- [47] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1998.
- [48] L. Itti and C. Koch, "Computational modelling of visual attention," *Nature reviews neuroscience*, 2001.
- [49] H. Larochelle and G. E. Hinton, "Learning to combine foveal glimpses with a third-order boltzmann machine," in *Conference on Neural Information Processing Systems*, 2010.
- [50] V. Mnih, N. Heess, A. Graves, and K. Kavukcuoglu, "Recurrent models of visual attention," in *Conference on Neural Information Processing Systems*, 2014.
- [51] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Conference on Neural Information Processing Systems*, 2017.
- [52] T. Bluche, "Joint line segmentation and transcription for end-to-end handwritten paragraph recognition," in *Conference on Neural Information Processing Systems*, 2016.





7. Experiments & Evaluation

7.1 Dataset

We used the following set of research papers from arXiv and Google Scholar as our evaluation dataset. These papers were selected because they contain dense technical content, strong citation structures, and diverse subject areas spanning NLP, computer vision, biomedical language models, and generative modeling. This diversity allowed us to observe how well our Graph-RAG system handled different writing styles, terminology, and citation patterns.

The dataset included the following papers:

- Attention Is All You Need
- Squeeze-and-Excitation Networks
- Non-local Neural Networks
- Dual Attention Network for Scene Segmentation

- BioBERT: a pre-trained biomedical language representation model for biomedical text mining
- BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension
- Transformers: State-of-the-Art Natural Language Processing
- Swin Transformer: Hierarchical Vision Transformer using Shifted Windows
- Masked Autoencoders Are Scalable Vision Learners
- High-Resolution Image Synthesis with Latent Diffusion Models
- Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks

7.2 Evaluation

Since this project focuses on building a functional hybrid Graph RAG architecture rather than developing new numerical benchmarks, we used general qualitative evaluation methods to assess system performance.

General Evaluation Approach

- **Relevance of Retrieved Chunks** – Checked whether the top retrieved segments semantically aligned with the user query.
- **Citation Awareness** – Verified whether graph traversal surfaced papers connected through citations, even when not semantically closest.
- **Consistency of LLM Output** – Ensured generated answers were coherent, grounded, and referenced the retrieved chunks.
- **Reduction of Hallucinations** – Observed if answers stayed within retrieved context.
- **User Transparency** – Reviewed the clarity of citation trails and chunk-level attribution.

Tracing-Based Validation

Using Phoenix tracing, we validated the full reasoning flow by:

- Inspecting LLM calls and inputs
- Verifying retrieval context passed to the LLM
- Reviewing model outputs for groundedness
- Checking cost, latency, and sequence of operations

The screenshot in the evaluation section demonstrates a full trace of the system handling an academic question. The trace verifies that:

- Retrieval pulled relevant context
- The LLM generated grounded explanations
- No unsupported citations were produced when papers did not exist in the dataset
- This form of trace-driven qualitative evaluation helped ensure system transparency and correctness even without formal quantitative metrics.

Trace Details

Trace Status: OK Total Cost: \$0 Latency: 15.9s

Trace

Component	Duration
RunnableSequence	15.9s
RunnableParallel<context,questi...	1ms
RunnableLambda	1ms
RunnableLambda	0ms
ChatPromptTemplate	0ms
ChatOllama @ 472	15.9s
StrOutputParser	1ms

RunnableSequence 15.9s at 12/05/2025, 11:25:11 AM

All Attributes

```

1 {
2   "input": {
3     "mime_type": "application/json",
4     "value": "{\n  \"question\": \"How is attention different from sequence models ?\",\n  \"context\": \"Paper 1: Deep Learning for Generic Object Detection: A Survey (2019, citations: 2594)\\n\\nPaper 2: Graph neural networks: A review of methods and applications (2020, citations: 4755)\\n\\nPaper 3: Ref W2970686063 (citations: 0)\\n\\nPaper 4: Ref W1991367009 (citations: 0)\\n\\nPaper 5: Ref W1502661168 (citations: 0)\\n\""}
5   },
6   "openinference": {
7     "span": {
8       "kind": "CHAIN"
9     }
10  },
11  "output": {
12    "value": "Here are concise bullet points explaining the difference between attention and sequence models:\n\n* **Attention Mechanism**: Attention is a mechanism used in neural networks to focus on specific parts of an input sequence when generating an output. It allows the model to weigh the importance of each element in the input sequence relative to others [1].\n\n* **Sequence Models**: Sequence models, on the other hand, are a class of machine learning algorithms that process sequential data, such as text or speech. They typically rely on recurrent neural networks (RNNs) or long short-term memory (LSTM) networks to capture temporal dependencies in the data.\n\n* **Key difference**: The key difference between attention and sequence models is that attention is a mechanism used within sequence models to selectively focus on specific parts of the input sequence, whereas sequence models are a broader class of algorithms that process sequential data.\n\nReferences:\n[1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.\n\nNote: Since Paper 3, Paper 4, and Paper 5 do not exist or are not available, I was unable to include citations from those papers."
13  }
14 }

```

8. Conclusion & Future Directions

8.1 Conclusion

The Graph RAG Powered Academic Assistant successfully demonstrates how combining semantic retrieval with citation graph traversal can significantly enhance the accuracy, reliability, and transparency of AI assisted academic research. Traditional RAG systems often struggle with hallucinations, shallow context interpretation, and the inability to trace the lineage of ideas across scholarly works. By integrating citation aware graph structures and implementing a hybrid retrieval mechanism, our system provides users with contextualized, well-grounded answers supported by verifiable sources.

Despite utilizing a relatively small locally hosted model (LLaMA 3.2B via Ollama), the system produced consistently strong, contextually grounded responses. The hybrid retrieval architecture amplified model performance by delivering highly relevant, citation-supported context to the LLM, effectively compensating for its smaller size. This confirms that thoughtful system design particularly retrieval strategy and context conditioning can significantly improve answer quality without relying on larger, more computationally expensive models.

The inclusion of explainability components such as source attribution, citation trails, and Phoenix-based LLM tracing further enhances user trust and academic reliability. These features ensure that each response can be traced back to specific chunks, citations, and reasoning steps, promoting transparency and supporting academic integrity. The streamlined interface also allows students and researchers to intuitively explore document relationships, retrieved evidence, and generated insights.

Overall, this project demonstrates the feasibility and value of a Graph RAG architecture for academic research applications. It establishes a strong foundation for future advancements in explainable, citation aware AI

systems, highlighting the importance of structured retrieval and interpretability in scholarly environments.

8.2 Future Directions

Building upon the strong foundation of the current Graph-RAG Powered Academic Assistant, several enhancements can significantly expand its capabilities, scalability, and real-world applicability:

- **Explore Different LLM Models:** A key direction for future enhancement is exploring a broader range of LLM models, including larger open-source architectures, instruction-tuned variants, and domain-specific fine-tuned models. Evaluating multiple LLM families will help determine which configurations produce the best retrieval accuracy, reasoning depth, and citation grounding within the hybrid Graph-RAG framework.
- **Modular Multi-Agent Architecture:** Transforming the current monolithic pipeline into a modular multi-agent system—such as Document Agent, Citation Agent, Retrieval Agent, and Reasoning Agent would improve scalability, enable parallel processing, and support integration with emerging multi-agent orchestration frameworks. This design could also enhance fault tolerance and make the system more adaptable for research or enterprise environments.
- **Cloud Deployment on AWS:** Deploying the system on AWS services such as S3, EC2/ECS, Lambda, API Gateway, and DynamoDB would allow the platform to scale to thousands of users. AWS-native architecture would enable persistent vector storage, distributed retrieval, serverless scaling, and secure integration with academic institutions. Future iterations may also adopt managed graph services for citation analysis.
- **Support Large-Scale Ingestion with Neo4j:** To handle large volumes of scholarly documents and dense citation networks, the system can integrate Neo4j or other graph databases. This would enable faster traversal, advanced analytics (e.g., PageRank, community detection),

and a more comprehensive citation ecosystem capable of supporting thousands of interconnected research papers.

- **Fine-Tune Domain-Specific LLMs:** Future versions of the system could incorporate fine-tuned LLMs trained on biomedical, legal, engineering, or educational corpora. Such models are better suited for nuanced scientific language, enabling more accurate interpretations of domain-specific terminology and improving reasoning quality.
- **Enhanced Web UI with Search Filtering & Citation Heatmaps:** An advanced frontend could introduce interactive research filtering, citation density heatmaps, topic clustering, paper relationships, and hop-based exploration tools. These features would make the system more visually intuitive and provide a richer exploratory experience for students and researchers.

9. References

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- **Hybrid RAG** - <https://arxiv.org/pdf/2408.04948>